Module 012 Reasoning Studio

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Abstract—This module forms a critical component of the CEREBRUM framework, a community-structured graph attention engine for Knowledge Graph reasoning. It focuses on Advanced Graph Attention Analysis.

Index Terms—Knowledge Graph, Graph Attention, DSCF, CSA, Hierarchical Reasoning.

1 Glass-Box Reasoning Studio: Visualizing Graph Attention and Latent Multi-Hop Inference

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1.0.1 Abstract

The "Black-Box" nature of modern Graph Neural Networks (GNNs) and Transformer-based reasoning systems limits their utility in domains requiring high auditability. We present the **Glass-Box Reasoning Studio**, an interactive visualization framework designed for the forensic audit of multi-hop Knowledge Graph inference. The Studio reifies the "Reasoning Beam" as a dynamic topological trace, where edges are scaled by their **Community-Structured Attention (CSA)** weights and nodes are color-coded by their **DSCF/TSC** community partitions. We introduce a "Forensic Score Breakdown" interface that exposes the latent mathematical signals (semantic similarity, community guidance, and structural centrality) driving each traversal hop, building on foundational Explainable AI (XAI) principles [**?**], [**?**], [**?**]. Furthermore, we describe a real-time "Live Feed" visualization for streaming graphs that animates **STDP spike events** and the materialization of speculative causal links. The v1.2.0 release adds adaptive node clustering to support visual scaling for graphs exceeding 10^5 nodes. Our results show that this interactive "Glass-Box" approach significantly reduces the time required for human experts to verify complex AI-generated hypotheses.

1.0.2 1. Introduction

Explainability in AI (XAI) has traditionally focused on post-hoc interpretations of neural weights (e.g., saliency maps). In graph-based reasoning, however, the explanation is the path itself. The Glass-Box Reasoning Studio provides the first integrated environment for visualizing graph attention as a physical, navigatable flow.

1.0.3 2. Forensic Visualization Methodology

1.0.3.1 2.1 The Reasoning Trace: The Studio implements a path-centric rendering algorithm that isolates the sub-graph involved in a specific query. The attention weight $a(u, v, k)$ is mapped to edge thickness and opacity, allowing the user to visually perceive the "narrowing of the beam" as the AI focuses on likely answers.

1.0.3.2 2.2 Modal Animations: For temporal and streaming data, the Studio utilizes high-frequency state updates:

- **Potentiation**: Edges being strengthened by LTP (SPEC_003) increase in saturation.
- **Drift**: Community boundaries shift smoothly using force-directed layouts to reflect modularity updates (SPEC_007).

1.0.4 3. Interactive Debugging (v1.2.0)

The Studio provides a "Dialectical reasoning" mode where users can manually adjust CSA parameters (α, β, γ) via sliders and observe the immediate physical shift in the reasoning beam, providing a "Human-in-the-Loop" (HITL) interface for hyperparameter tuning. In v1.2.0, this includes real-time feedback submission to the **MetaParameterLearner**.

1.0.5 4. Conclusion

The Glass-Box Reasoning Studio transforms graph attention from an abstract mathematical construct into a tangible, auditable artifact. By bridging the gap between latent semantic operations and human-readable topologies, it enables the deployment of autonomous reasoning systems in high-stakes environments.

— **References**

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